

# Dimension Reduction Analysis: pbmc

Generated report #1

Seed: 0 | Plan revision: 2

## Dataset profile & preprocessing

Peripheral blood mononuclear cells (PBMCs) from a healthy donor, profiled with 10x Genomics single-cell RNA sequencing (scRNA-seq). This is the standard "pbmc3k" dataset used throughout the Scanpy tutorials and widely used as a benchmark for single-cell analysis methods. Stored as the raw h5ad file from the Scanpy tutorials.

Samples: 2,700    Features: 32,738    Sparsity: 97.4%

Missing values: No

### Preprocessing: `normalize_total`

Modality: sparse count / omics-like. `profile.json` shows 97.4% zeros, non-negative integer values in `[0, 419]`, and `DATA_DESCRIPTION.md` says these are raw, unnormalized UMI counts with total counts per cell varying widely (mean ~2,367; genes detected per cell 212 to 3,422). Rescaling each cell to a common total (default target: the median cell total) removes this sequencing-depth confound so cells are compared by expression profile, not by how deeply they were sequenced. `DATA_DESCRIPTION.md` states no normalization was applied upstream, so this is not double-normalizing.

### Preprocessing: `log1p`

Modality: sparse count / omics-like. After depth normalization the counts are still heavy-tailed (raw max 419 against a mean of 0.072 and std 1.129 per `profile.json`; `DATA_DESCRIPTION.md` warns a few highly-expressed genes dominate raw totals). `log(1+x)` compresses that tail so a handful of very high-count genes do not dominate every Euclidean distance, while keeping zeros at zero.

### Preprocessing: `select_top_variance`

Modality: wide, on top of count-like. 32,738 genes against 2,700 cells (features exceed samples about 12-fold), and `DATA_DESCRIPTION.md` notes many genes are detected in very few cells, so most columns carry mostly dropout noise. Keeping the 2,000 highest-variance genes after `normalize_total` + `log1p` cuts the  $O(N^2)$  methods' feature cost and their noise. Judgment call: 2,000 is the skill's `min(2000, n_features)` heuristic; alternatives were 1,000 (risks dropping lower-variance marker genes) or keeping all genes (noise-dominated distances). The step ranks by raw variance, not mean-adjusted dispersion, so some high-mean genes are favored. No standardize step afterwards: `data-profiling/SKILL.md` documents that on a matrix of this exact shape it changed fine neighborhoods substantially with no labels to show it helped.

## Methods and hyperparameters

### **pca (role: general\_purpose, status: reused)**

Hyperparameters: `n_components=50`

Required linear baseline and the trustworthy numeric representation used as the clustering source. `n_components = min(50, floor(2700/2)=1350, 2000) = 50`; with 2,000 retained genes and no labels, 50 components are enough to capture the main axes of variation among 2,700 cells without fitting individual-cell noise. The scree plot will show how concentrated the variance is.

### **kernel\_pca (role: general\_purpose, status: reused)**

Hyperparameters: `n_components=50, kernel=rbf, gamma=0.0017`

The required PCA variant. Kernel PCA is feasible because `n_samples = 2,700` is below the ~5,000 threshold for its  $N \times N$  kernel. Judgment call on `gamma`: sklearn's default `1/n_features = 0.0005` gives a kernel value of about  $\exp(-0.0005 * 577) = 0.75$  at the median pairwise squared distance (577, measured on a 2,000-cell fixed-seed subsample of the preprocessed matrix), so nearly every pair would look similar and the kernel would behave almost linearly, reproducing PCA. `gamma = 0.0017` (about  $1/\text{median squared distance}$ ) puts a typical pair at  $\exp(-1)$ , so the kernel can pick up nonlinear structure. Alternatives: the default (near-linear), or a much larger `gamma` (kernel collapses toward the identity and becomes noise-dominated). `n_components = 50` matches PCA for a like-for-like comparison.

## Methods and hyperparameters (cont.)

### **diffusion\_maps (role: general\_purpose, status: ok)**

Hyperparameters: `n_components=10`, `alpha=1.0`

A second nonlinear general-purpose embedding, chosen because PBMC scRNA-seq cell states can form continuous gradients that a linear projection flattens; `n_samples = 2,700` keeps its  $N \times N$  affinity matrix affordable. `epsilon` is left at the implementation's default (median pairwise squared distance, about 577 on this preprocessed matrix). Revision 2, judgment call: the critic noted that at this bandwidth the eigenvalues (0.107, 0.034, 0.024, ...) track PCA's variance ratios, so the method mostly reproduces PCA. A local bandwidth, `epsilon = 125` (one third of the measured median squared distance to the 15th nearest neighbor, 375), was tried: its eigenvalues rose to 0.79-0.53 but the leading eigenvectors localized on about a dozen outlier cells and collapsed the other cells to one point, which is a degenerate embedding. The cause is distance concentration: the 1st and 15th nearest neighbors (median squared distance 342 and 375) are not much closer than a typical pair (577), so no single bandwidth is both local and well-connected. The default was restored and the method is reported as largely reproducing PCA, not as an independent nonlinear view. `alpha = 1.0` applies density normalization so the geometry is not driven by how many cells sit in each region (data-derived cluster sizes differ about 4-fold, 1,330 to 337). `n_components = 10` because the eigenvalues fall from 0.107 to 0.0047 by the 10th.

### **umap (role: visualization\_only, status: reused)**

Hyperparameters: `n_components=2`, `n_neighbors=15`, `min_dist=0.1`, `random_state=0`

Visualization only, for a 2D picture of the 2,700 cells; never an input to clustering or any other computation. `n_neighbors = 15`: the smallest density group found in revision 1 had 347 cells and the smallest k-means cluster 337, so a 15-cell neighborhood is well under 5% of the smallest group and can resolve it without merging it into a neighbor, while still large enough to connect the graph across 2,700 cells. `min_dist = 0.1` keeps tight groups compact, which matters because this layout is also the density cross-check reference for choosing `k` (DBSCAN nominates candidate groups here; they must then be separated by k-means on PCA). A larger `min_dist` would spread groups out and blur density gaps.

## Methods and hyperparameters (cont.)

### **tsne (role: visualization\_only, status: reused)**

Hyperparameters: `n_components=2, perplexity=30, random_state=0`

A second visualization-only layout. CLAUDE.md requires two seeds or settings before any structural claim from a manifold picture, and artifacts are keyed by method name, so a second method (not a second UMAP run) is the available check. There are no labels, so agreement between UMAP and t-SNE on which groups separate is the only internal consistency check. `perplexity = 30` is an effective neighborhood of about 30 cells, under 10% of the smallest group (337 to 347 cells), so it targets the same local scale as UMAP's `n_neighbors = 15` and can resolve every group large enough to matter for the coloring. Limitation: only one seed per method.

### **Data-derived clusters (k-means on pca; colors only)**

`k=4` was chosen by a density cross-check, not silhouette alone: DBSCAN on the umap visualization (`eps=0.2178`, auto-selected via the k-distance knee, `min_samples=10`) found 3 groups of size [1613, 679, 347] well above noise; `k=4` is the smallest `k` in 2-12 where every one of those groups lands in its own k-means cluster on the pca embedding (silhouette `k=2: 0.365`, `k=3: 0.304`, `k=4: 0.289`, `k=5: 0.256`, `k=6: 0.204`, `k=7: 0.206`, `k=8: 0.142`, `k=9: 0.129`, `k=10: 0.102`, `k=11: 0.123`, `k=12: 0.115`). The visualization was used only to nominate candidate groups, never as an input to the clustering itself. Cluster sizes: 1,330, 685, 348, 337. These clusters only color the plots of an unlabeled dataset; they are a description of the embedding, not a finding, and no biological or domain meaning is attached to them.

`profile.json` has `has_labels: false` and `DATA_DESCRIPTION.md` confirms `adata.obs` is empty, so without this block every plot would be a single-color density scatter. k-means on the 50-component PCA embedding (general-purpose, never on UMAP or t-SNE) gives data-derived clusters used only to color the plots. Judgment call on `k_range = [2, 12]`: the density cross-check selects the smallest `k` that separates every DBSCAN candidate group, and a k-means split can need more clusters than there are groups, so the range needs room above the group count; 12 keeps the smallest cluster at roughly  $2,700 / 12 = 225$  cells on average, above UMAP's 15-cell neighborhood. The alternative `[2, 10]` would also have worked here, since `k = 4` was selected. These clusters are not validated against anything and are not named as cell types.

# Quantitative metrics

| method         | trustworthiness | variance explained<br>(PCs used) | PCs for 90%<br>variance |
|----------------|-----------------|----------------------------------|-------------------------|
| pca            | 0.9344          | 0.2904                           | >200                    |
| kernel_pca     | 0.8824          |                                  |                         |
| diffusion_maps | 0.8466          |                                  |                         |
| umap           | 0.7740          |                                  |                         |
| tsne           | 0.8023          |                                  |                         |

## How to read the metrics

### Trustworthiness (`sklearn.manifold.trustworthiness`, 10 neighbours)

For each sample, takes its 10 nearest neighbours in the embedding and penalizes any that were not close neighbours in the original preprocessed data, weighting each penalty by how far down the original neighbour ranking that point sat. Ranges 0 to 1; 1 means the embedding introduces no false neighbours. It measures local neighbourhood fidelity only, not preservation of global distances. Computed against the data after the plan's preprocessing steps; on a random 5,000-sample subset when the dataset is larger than that (see Limitations).

### Variance explained (PCs used)

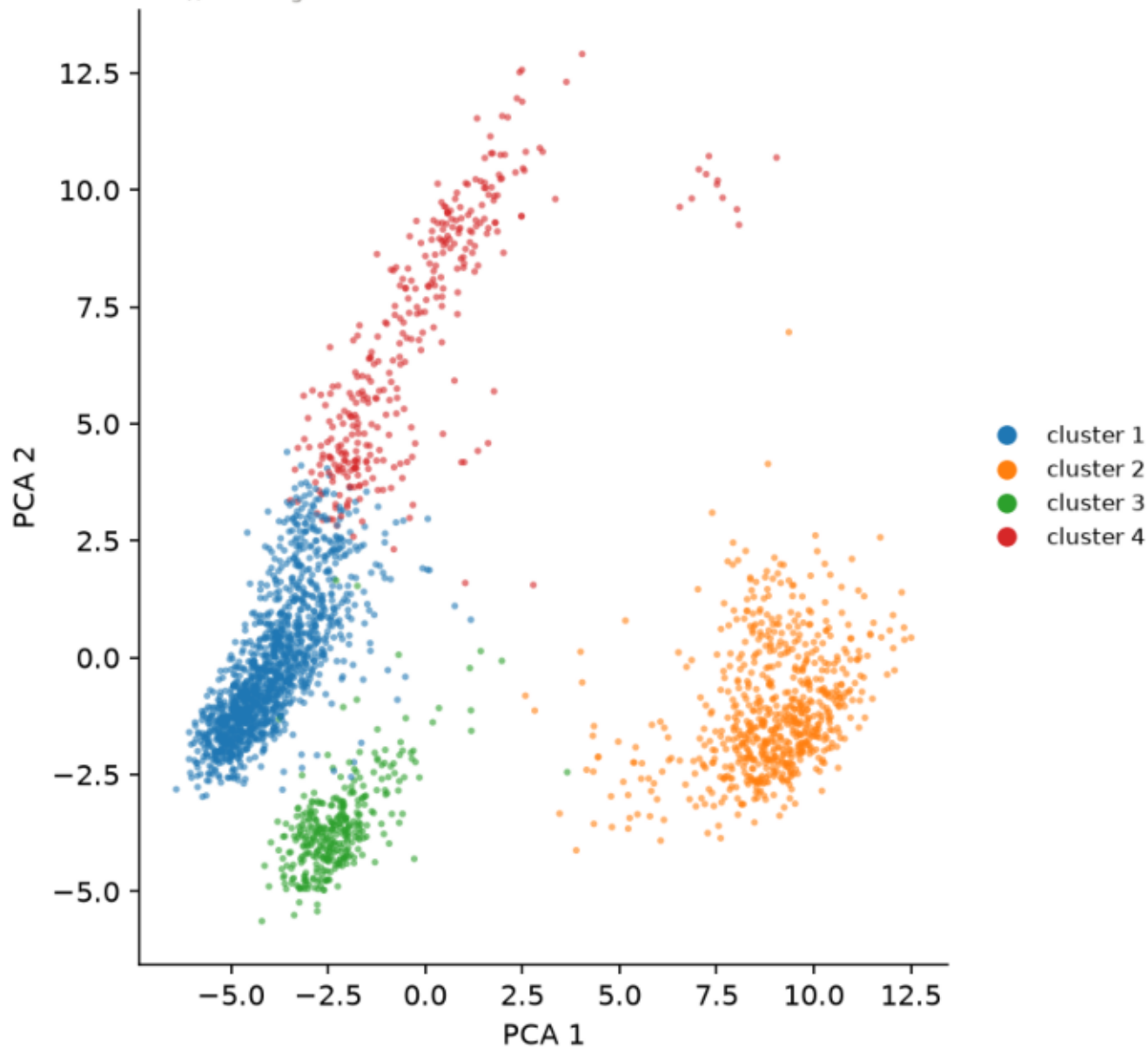
The fraction of the total variance in the preprocessed data captured by the components actually kept in the embedding. For PCA, the sum of those components' explained-variance ratios (each component's variance divided by the total). For Sparse PCA, whose components are not orthogonal, no such per-component decomposition exists; instead this is the variance captured by the subspace the components span (the components orthogonalized via QR, then variance summed along that orthonormal basis) – a valid but different construction, marked in the sidecar JSON. Ranges 0 to 1. Blank for methods neither applies to.

### PCs for 90% variance

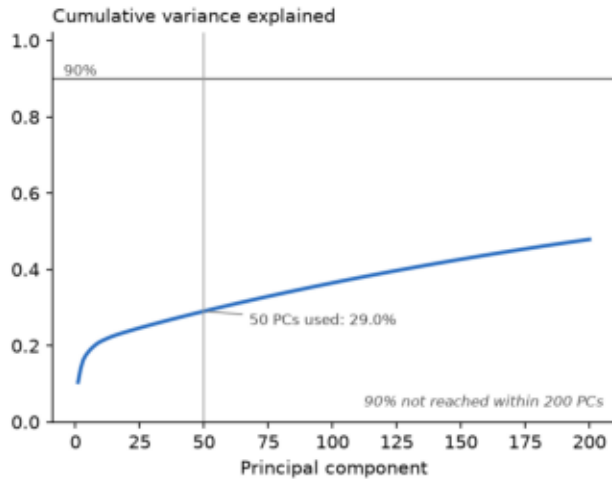
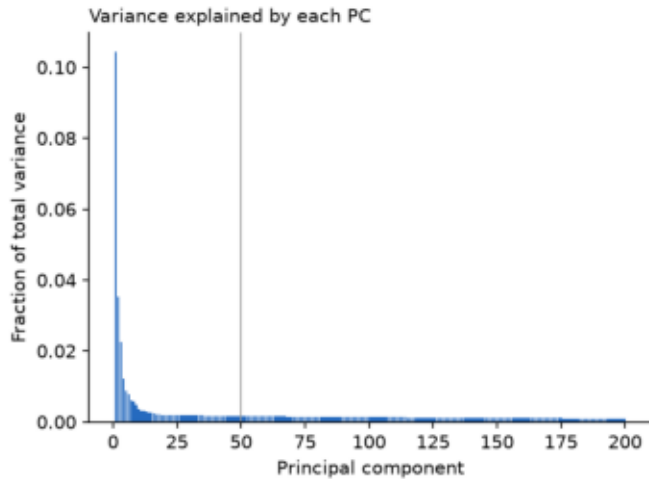
The smallest number of components whose cumulative explained variance reaches 90%. For PCA, read off a diagnostic fit of up to  $\min(4 \times n_{\text{components}}, 200)$  components, deeper than what the embedding uses, so this can exceed the PCs actually kept. For Sparse PCA there is no deeper fit (it is too expensive to refit just for diagnostics); this is bounded by the components actually requested. Either way, a leading '>' means 90% was not reached within what was fit, so the true number is larger than shown. The scree plots show the full curve.

## pbmc: pca

n=2,700, colored by data-derived clusters (k-means on pca, k=4), showing dims 1-2 of 50

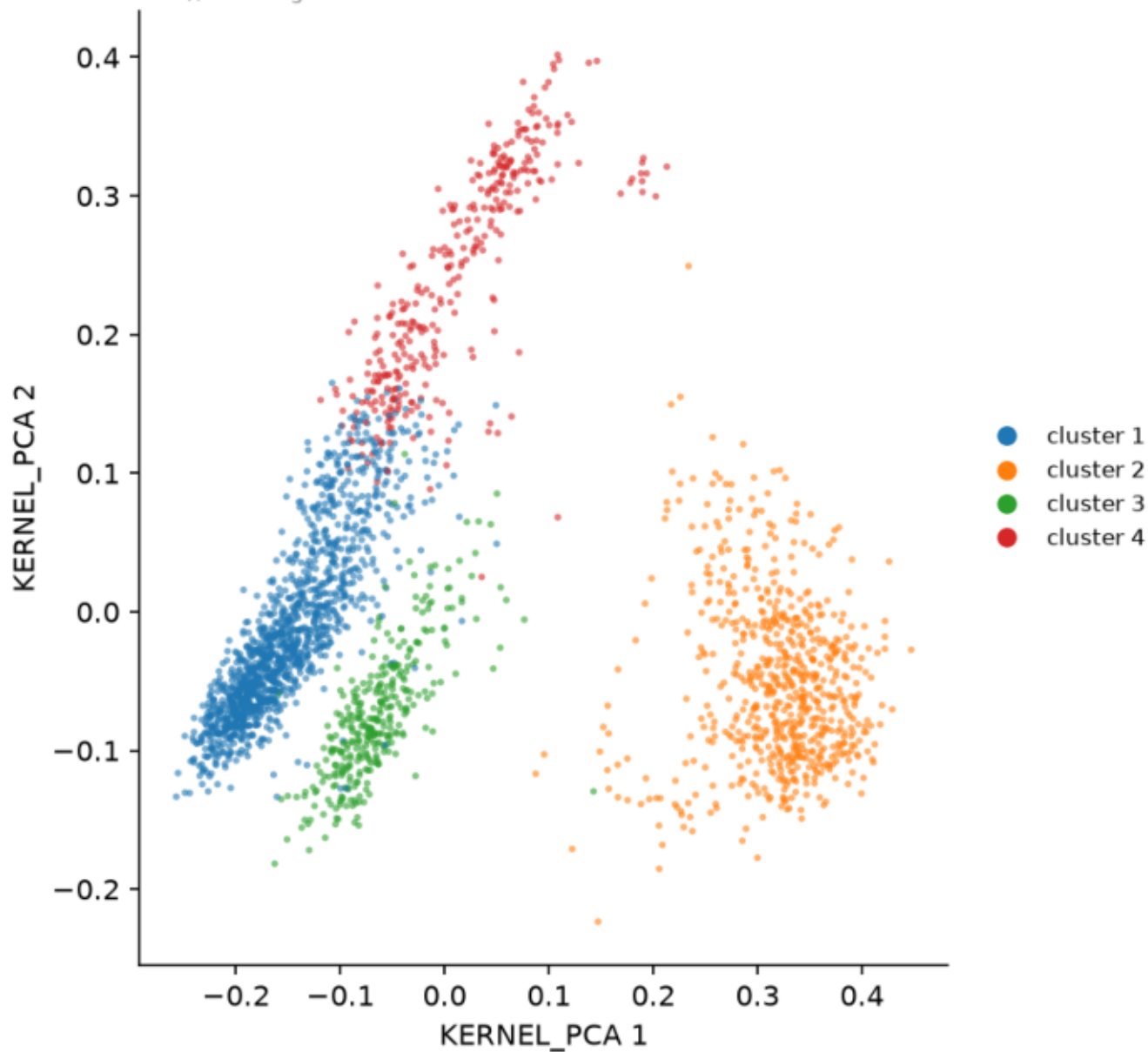


pbmc: pca scree plot



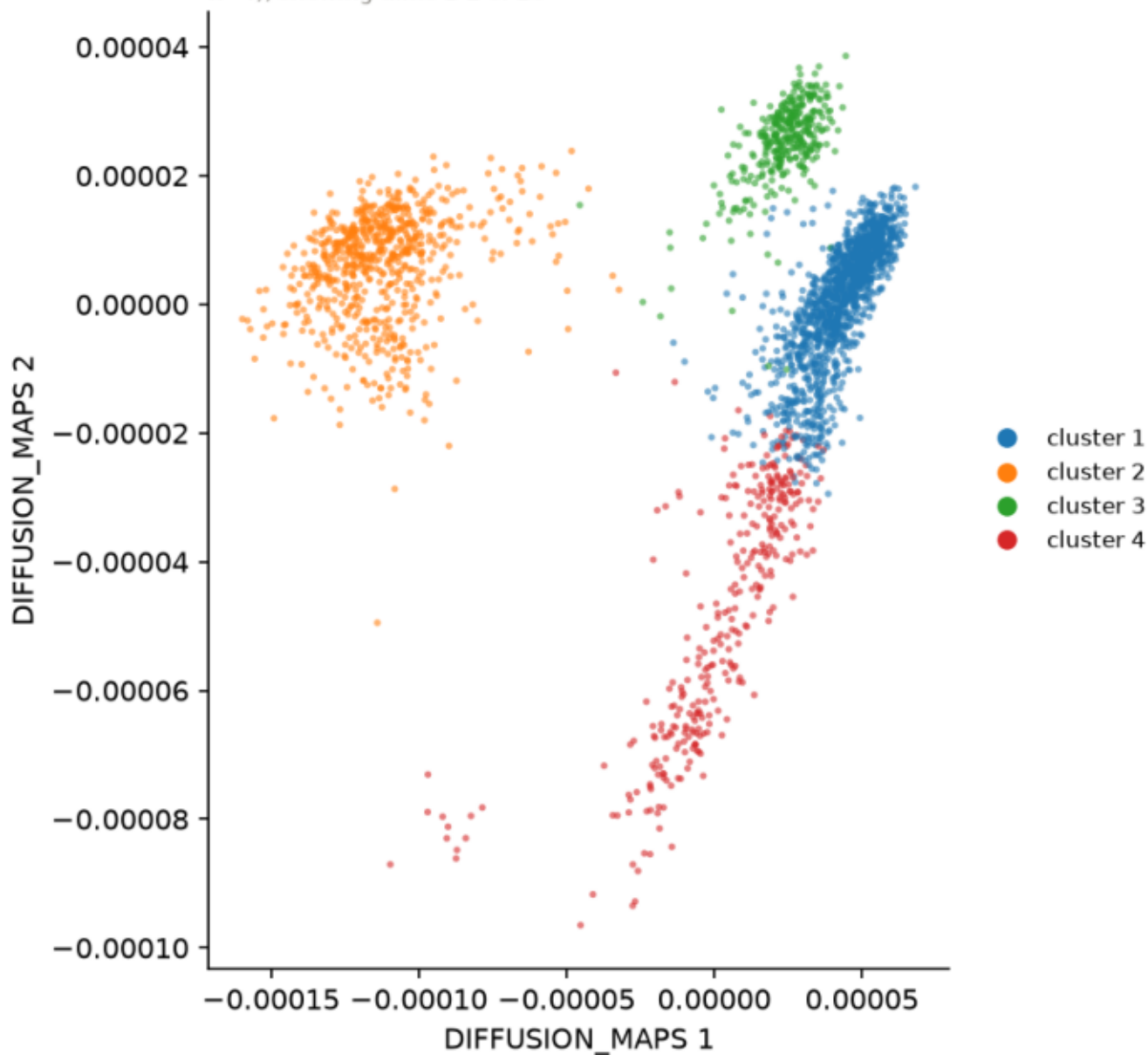
# pbmc: kernel\_pca

n=2,700, colored by data-derived clusters (k-means on pca, k=4), showing dims 1-2 of 50



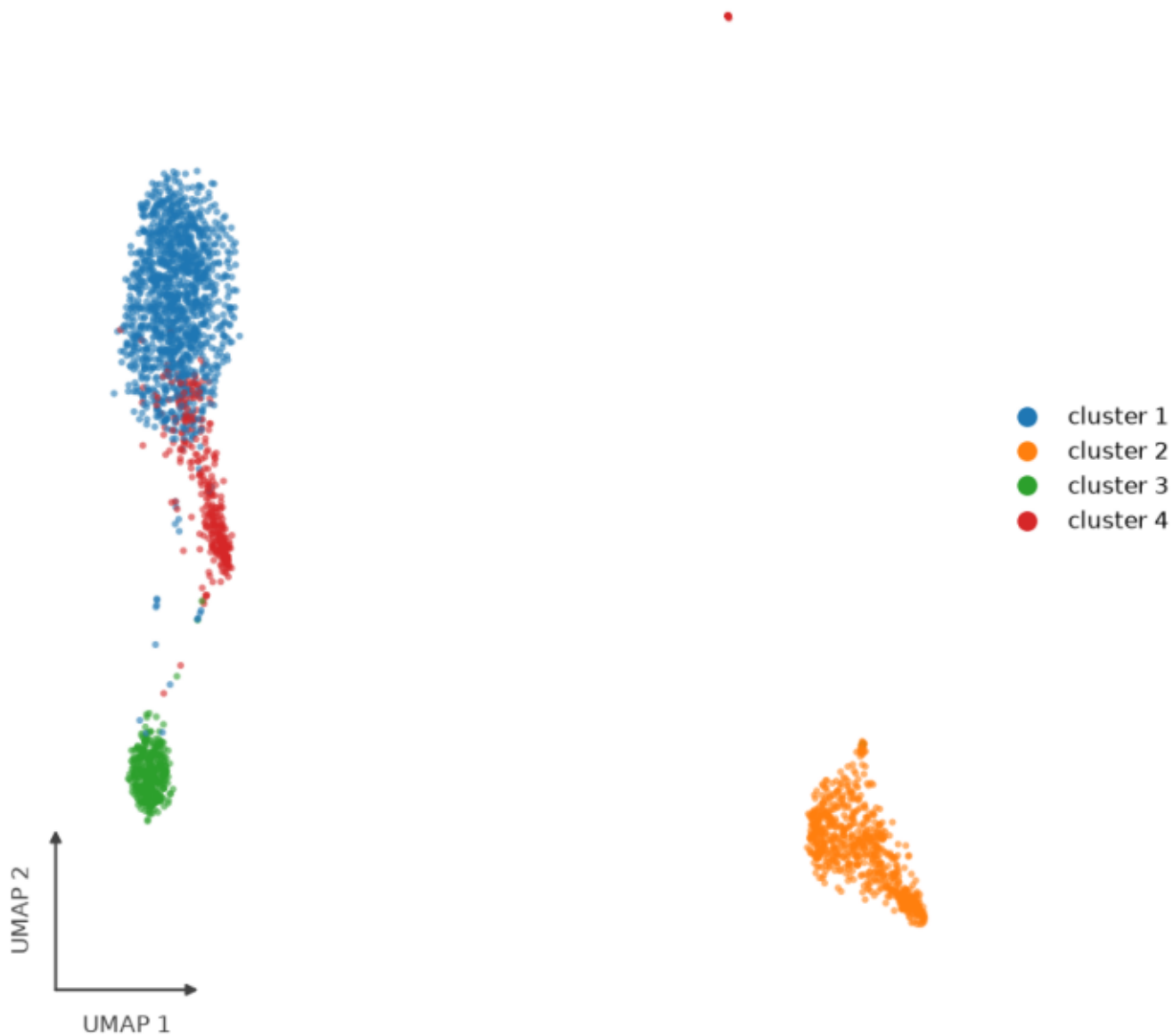
# pbmc: diffusion\_maps

n=2,700, colored by data-derived clusters (k-means on pca, k=4), showing dims 1-2 of 10



## pbmc: umap

n=2,700, colored by data-derived clusters (k-means on pca,  
k=4)

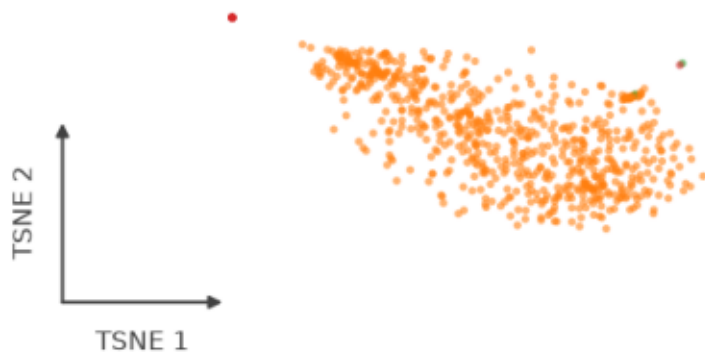


## pbmc: tsne

n=2,700, colored by data-derived clusters (k-means on pca,  
k=4)



- cluster 1
- cluster 2
- cluster 3
- cluster 4



# Findings and interpretation

## What each method produced

All five methods in the plan ran successfully on the same preprocessed matrix: 2,700 cells by the 2,000 highest-variance genes, after depth normalization and log1p. Every stochastic step used seed 0.

PCA (50 components) explains 0.2904 of the total variance. The first three components explain 0.1044, 0.0354 and 0.0226. After roughly the first 10 components, each further component adds only a small, nearly constant share. Within the 200-component diagnostic fit, 90% of the variance was not reached, so the number of components needed for 90% is >200. The scree plot shows a few strong directions followed by a long, flat tail. Most of the variance in this 2,000-gene space is spread thinly across many directions, which is what you expect when per-gene noise such as dropout makes up much of the signal.

Kernel PCA (RBF, gamma = 0.0017, 50 components) has trustworthiness 0.8824, compared with 0.9344 for 50-component PCA. Its first two dimensions look very similar to PCA's, with the same four colored regions in the same arrangement. So at this bandwidth, the nonlinear kernel did not reveal a clearly different large-scale layout.

Diffusion Maps (10 components, default bandwidth) has trustworthiness 0.8466. Its eigenvalues (0.1069, 0.0342, 0.0244, ...) closely track PCA's variance ratios, and its first two dimensions look like a reflected version of the PCA plot. The critique flagged this. A more local bandwidth was tried in revision 2, but it gave a degenerate embedding, with the leading dimensions dominated by a handful of outlier cells, so the default was restored. Here, Diffusion Maps should be read as largely repeating PCA, not as an independent nonlinear view.

UMAP (n\_neighbors = 15, min\_dist = 0.1) has trustworthiness 0.774 and t-SNE (perplexity = 30) has 0.8023, both in 2 dimensions.

These trustworthiness values come from embeddings of different dimensionality (50, 50, 10, 2 and 2). They do not rank the methods. A 50-dimensional embedding has much more room to keep each cell's 10 nearest neighbors than a 2-dimensional one.

## Findings and interpretation (cont.)

### What the figures show

All plots are colored by data-derived clusters: k-means with  $k = 4$  on the 50-component PCA embedding.  $k$  was chosen by the density cross-check. DBSCAN on the UMAP layout nominated 3 candidate groups (1,613, 679 and 347 cells), and  $k = 4$  was the smallest  $k$  in the scanned range  $[2, 12]$  that put each of them in a different k-means cluster. The k-means clusters hold 1,330, 685, 348 and 337 cells. The silhouette at  $k = 4$  is 0.289. It was higher at  $k = 2$  (0.365), but that split would merge groups the density check separates. These clusters are only a coloring. They are not validated against anything and are not cell types.

The two visualization layouts agree on coarse structure. Both show cluster 2 as a large group far from the rest, cluster 3 as a compact, separate group, and clusters 1 and 4 as touching regions with no clear gap between them. In UMAP, cluster 4 looks like an extension of cluster 1 rather than a separate island. In PCA and Kernel PCA, cluster 4 lies along a direction running away from cluster 1. The same three-region pattern shows up in all four general-purpose and visualization plots. That is consistent evidence for about three well-separated regions plus one gradient-like region, but it is exploratory, not confirmed.

A very small group of cells sits apart from everything else in every embedding (top of the UMAP plot, upper right of the PCA plot). It is colored as part of cluster 4, but it does not belong with the rest of that cluster. It is too small to register as a group in the density check, which uses DBSCAN with `min_samples = 10` and a minimum group size, so the coloring does not capture it. It may be a real small population or a group of low-quality cells. The data cannot say which.

### How the methods relate

PCA is the trustworthy numeric summary here and the input to the clustering. Kernel PCA and Diffusion Maps were included as nonlinear general-purpose alternatives. In this run, both ended up close to PCA rather than showing new structure. UMAP and t-SNE are pictures for a human reader only. Their cluster sizes, shapes and between-cluster distances are not findings, and nothing downstream used them as input. Their main role here is the cross-check: two independent layouts show the same coarse grouping.

## Findings and interpretation (cont.)

### What can't be concluded

There are no labels, so no cluster can be named or checked against a known cell type, and silhouette on data-derived clusters only measures how cleanly k-means split the PCA space. Each manifold method was run with one seed, so fine details of either layout (the shape of cluster 1, the gradient in cluster 4) have only a single run behind them. The analysis applied no quality-control filtering. DATA\_DESCRIPTION.md notes that low-quality cells and doublets may be present, and some visible structure, including the small isolated group, could reflect cell quality rather than biology. Finally, the scree plot's flat tail means the nonlinear methods worked on distances in which much of the variation is spread-out noise. That is the most likely reason Kernel PCA and Diffusion Maps added little beyond PCA.

## Critique and revision

Reviewed plan revision 1. Readiness: revise: at least one finding was addressed with a plan revision.

6 finding(s): critical=0, moderate=2, minor=4

### F1: instability\_or\_metric (moderate)

Diffusion maps appears to behave almost like PCA, not like the nonlinear trajectory-preserving embedding the plan's reason expects, most likely because its default bandwidth is too wide.

Evidence: Plan reason: diffusion maps is chosen because PBMC 'cell states can form continuous trajectories that a linear projection flattens', with epsilon 'left at the implementation's data-driven default (median pairwise squared distance)'.  
diffusion\_maps.json eigenvalues are [0.1069, 0.0342, 0.0244, 0.0131, 0.0111, ...], which closely track pca.json explained\_variance\_ratio [0.1044, 0.0354, 0.0226, 0.0123, 0.0088, ...]. For a diffusion operator on data with real multi-scale structure, the leading non-trivial eigenvalues are normally close to 1; values near 0.1 mean the random walk mixes almost immediately, which is what a nearly uniform kernel produces. The kernel\_pca reason itself measured that at the median squared distance (about 577)  $\exp(-d^2/\text{median}) = \exp(-1)$ , so most pairs get similar affinity. The coordinates in diffusion\_maps.png are on the order of  $1e-5$ , and the layout looks like a reflected copy of pca.png (same elongated red arm, same small offshoot island, same isolated orange group). Trustworthiness is 0.8466, below PCA's 0.9344.

Recommendation: Set diffusion maps' bandwidth explicitly and give a reason, for example a local k-nearest-neighbor bandwidth or a small fraction of the median squared distance, and check that the leading eigenvalues move toward 1. If the implementation does not expose epsilon, keep the method but say in the report that at this bandwidth it mostly repeats PCA and gives no independent nonlinear view.

## Critique and revision (cont.)

### F2: preprocessing\_mismatch (moderate)

The shared preprocessing ends at 2,000 genes with no denoising step, so the neighbor- and kernel-based methods (kernel PCA, diffusion maps, UMAP, t-SNE) and their trustworthiness scores all work on distances that the metrics suggest are mostly noise.

Evidence: `pca.json`: the first 50 PCs explain only 0.2904 of the variance, and 90% is not reached within 200 PCs (`components_needed_for_90pct_variance_is_lower_bound: true`). The per-component ratio falls to a flat floor of about 0.0016 to 0.002 from roughly PC 20 onward (`pca_scree.png`), which is the usual signature of isotropic noise. So most of each pairwise Euclidean distance in the 2,000-gene space comes from that flat tail. The preprocessing list is `[normalize_total, loglp, select_top_variance(2000)]` with no reduction step before the nonlinear methods. This fits the low neighborhood scores for the 2D layouts (umap 0.774, tsne 0.8023) and the near-uniform kernel in F1. The structure itself is fine: there is exactly one shared preprocessing block, applied to every method.

Recommendation: If the preprocessing vocabulary supports it, add a shared PCA-denoising step after `select_top_variance` (for example about 20 to 50 components), with a reason based on where the scree tail flattens, so every method sees the same denoised input. Then recompute kernel PCA's gamma, because the measured median squared distance of 577 would change. If no such step is available, state in the report that the nonlinear methods and their trustworthiness were computed on a 2,000-dimensional space in which roughly 70% of the variance is spread thinly across the tail components.

## Critique and revision (cont.)

### F3: unjustified\_hyperparameter (minor)

Several diffusion maps settings have no reason specific to this dataset, or no reason at all.

Evidence: `alpha = 1.0` is in the hyperparameters but the reason never mentions it. `n_components = 10` is justified only as '10 components, since its role is a general-purpose representation, not a 2D picture', which would read the same on any dataset. The epsilon choice is justified as 'the implementation's data-driven default ... which adapts to this matrix's scale', which is also generic, and F1 suggests it does not suit this matrix.

Recommendation: Give alpha a reason (for example, `alpha = 1` removes the effect of sampling density, which matters if the population sizes differ a lot, as the cluster sizes 1330/685/348/337 suggest). Tie `n_components` to the eigenvalue decay once the bandwidth is fixed. Replace the epsilon reason with one that cites a measured distance scale.

### F4: unjustified\_hyperparameter (minor)

The UMAP and t-SNE neighborhood settings rest only on a sample-size bracket, and UMAP's `min_dist` has no reason.

Evidence: UMAP reason: '`n_neighbors = 15` because `n_samples = 2,700` falls in the 500 to 20,000 range, where 15 balances local detail against global coherence; `min_dist = 0.1`.' The `min_dist` value is stated but not justified. t-SNE reason: '`perplexity = 30` is small relative to 2,700 cells, so it emphasizes local neighborhoods'. Both would read the same on any dataset of 500 to 20,000 samples and do not cite anything about this data, such as the smallest group size the plan expects to resolve. This matters because the DBSCAN cross-check found groups as small as 347 cells, and there is a visible group of about 10 cells (F5).

Recommendation: Tie `n_neighbors` and `perplexity` to the smallest group size the layouts need to resolve, and give `min_dist` a reason (for example, a low `min_dist` keeps tight groups visually compact for the density cross-check). Because the plan uses one seed per manifold method, consider a second `n_neighbors` or `perplexity` setting as the guard-rail cross-check instead of relying only on UMAP-vs-t-SNE agreement.

## Critique and revision (cont.)

### F5: instability\_or\_metric (minor)

A small, well-separated group of about 10 cells shows up as a separate island in all five embeddings, but it is colored as part of cluster 4, whose main body is elsewhere. The density cross-check could not have found it at `min_samples = 10`.

Evidence: The island is at about (7.5, 10.5) in `pca.png`, about (0.2, 0.31) in `kernel_pca.png`, about (-9e-5, -8e-5) in `diffusion_maps.png`, as the isolated red point at the top of `umap.png`, and as the isolated red point left of the orange group in `tsne.png`. `clustering.json`: `k_selection.min_samples = 10`, and `group_sizes [1613, 679, 347]` add up to 2,639 of 2,700 cells, so 61 cells were DBSCAN noise. A group about the size of `min_samples` cannot reliably be nominated. I am not recommending a different `k`. `majority_cluster_per_group` shows that the three nominated groups land in three distinct clusters, so the recorded evidence supports `k = 4` for those groups. The problem is that this fourth structure was never tested.

Recommendation: No hyperparameter change is required. The report should not call cluster 4 a single coherent group, and it should mention the small island as consistent across all five embeddings but not captured by the coloring. If the clustering block exposes the DBSCAN `min_samples`, lowering it with a stated reason would let the cross-check consider this group.

### F6: label\_leakage\_in\_reasoning (minor)

The clustering `k_range` upper bound is justified with outside knowledge of how many PBMC populations exist and with the visual outcome of an earlier run, not with this run's profile.

Evidence: Clustering reason: 'The upper bound of 12 (not 10) is a judgment call: PBMC samples are known to contain many immune populations, some small, and README.md records that an earlier pbmc run's silhouette-only choice picked `k = 2` and under-colored visible structure.' This is the same pattern as setting `k` from 'N known classes', although weaker: it only widens the search range, and the selected `k = 4` came from the density cross-check, not from this prior. The practical impact is small.

Recommendation: Rewrite the reason so the upper bound is justified by data-side facts, for example leaving the density cross-check room above the number of DBSCAN groups, or a bound based on `n_samples`, and drop the appeal to known PBMC population counts and prior-run appearance.

## Critique and revision (cont.)

### Critic's notes

All requested inputs were provided (plan.json revision 1, six metrics files, six figures). Checks that passed: (1) Manifold overread: two visualization\_only methods (UMAP seed 0, t-SNE seed 0) give independent looks. They agree at a coarse level: one large isolated group (orange), one compact separate group (green), and blue/red touching each other. Claims beyond that coarse level have only these two single-seed layouts behind them. (3) Structure: exactly one shared preprocessing list. PC1 at 0.104 against PC2 at 0.035 does not suggest an unaddressed depth confound, because normalize\_total is present. (4) The kernel PCA gamma reason is specific and well grounded (measured median squared distance of 577). A caution for the report: the trustworthiness values compare embeddings of different dimensionality (PCA 50-D 0.9344, kernel PCA 50-D 0.8824, diffusion maps 10-D 0.8466, t-SNE 2-D 0.8023, UMAP 2-D 0.774), so they should not be read as a ranking of methods. Also, clustering.json's cluster\_sizes and majority\_cluster\_per\_group may not share one index order: group 1 (679 cells) maps to cluster index 3, and if cluster\_sizes were indexed by label that cluster would hold only 337 cells, which is impossible. The report should not pair those two fields by index without checking.

### Revision applied (now revision 2)

dr-critic (revision 1) raised 6 findings. F1: an explicit local diffusion\_maps bandwidth (epsilon = 125) was tried; it produced a degenerate embedding (leading eigenvectors localized on ~12 outlier cells), so the default bandwidth was restored and the report states that diffusion maps largely reproduce PCA here. F3: added reasons for alpha and n\_components. F4: UMAP/t-SNE neighborhood sizes now tied to the smallest group size, and min\_dist justified. F6: clustering k\_range reason rewritten using only data-side facts. F2 (no shared PCA-denoise step exists in the preprocessing vocabulary; it is also the root cause of the F1 outcome) and F5 (a ~10-cell island below DBSCAN min\_samples = 10, which the clustering block does not expose) have no plan-level fix and are stated as limitations in the report.

## Limitations

No ground-truth labels available for this dataset; embedding quality assessed only via label-free trustworthiness, no supervised sanity check possible.

The cluster colors on unlabeled plots come from k-means ( $k=4$ ) on the pca embedding, with  $k$  picked by a density cross-check against a visualization, not validated against any ground truth. Different  $k$ , a different algorithm, or a different source embedding would color the same points differently; treat the colors as a visual aid, not as groups that are known to exist. Even the density cross-check only tests whether k-means keeps DBSCAN's candidate groups separate; it cannot confirm those candidate groups are meaningful in any domain sense.